

Engine Condition Prediction from Sensor Data

An MLP Classifier for Binary Engine Health Prediction | Exam 3 — Own ML Project

1. Problem Relevance

This project predicts binary Engine Condition (0/1) from six live engine sensor readings (RPM, lubricating oil pressure, fuel pressure, coolant pressure, lubricating oil temperature, coolant temperature) — the automotive counterpart to industrial predictive maintenance, using sensor channels comparable to standard OBD-II telemetry. The relevant stakeholders are fleet operators, telematics services, and ultimately driver-facing dashboard warnings: detecting engine problems from sensor drift before a breakdown or safety incident is a directly actionable use of classification. Critically, a false negative (predicting “fine” when the engine is actually in trouble) is a more serious failure mode than a false positive — this asymmetry motivates the per-class, threshold-aware evaluation approach used throughout Section 5 rather than reporting accuracy alone.

2. Data Choice

Source: an automotive engine-health / predictive-maintenance dataset (Kaggle-family; exact original authorship and collection methodology are undocumented — see Section 6). 19,535 rows × 7 columns (6 features + binary label), one row per sensor snapshot.

This dataset was selected over an earlier-considered 500k-row synthetic factory-sensor dataset because that dataset's Operational_Hours feature alone correlated at $r=-0.985$ with the target, leaving little room for a nonlinear model to add value. Here, by contrast, every individual feature correlates weakly with the label ($|r| < 0.27$; audit below), so the label depends on a genuinely multivariate, nonlinear combination — a setting where an MLP has real headroom to outperform a linear baseline.

Data Quality Audit

Check	Result	Consequence
Missing values	None (7/7 columns)	No imputation needed
Duplicate rows	None	No deduplication needed
Physically impossible values	None (no negative pressures/temps)	No hard cleaning needed
Class balance	63% class 1 / 37% class 0	Motivates class-weighted loss (§3)
Feature–label correlation	All weak: Engine rpm strongest ($r=-0.27$)	No single dominant feature; genuinely multivariate
IQR outlier scan	lub oil temp flags 13.4% (tight natural range, not errors)	No rows removed
Extreme value	1 row: Coolant temp = 195.5°C	Kept — plausible rare overheating event
Label semantics	Undocumented which of 0/1 = healthy	Reported symmetrically per class (§5, §6)

Expected Limitations

- No documented label semantics or collection methodology (real fleet telemetry vs. bench rig vs. partially synthetic is unknown).
- Only 6 sensor channels; production OBD-II systems expose more (throttle position, O2 sensor, battery voltage, etc.).
- 19,535 rows is workable but modest for a 6-feature nonlinear model — makes regularization a load-bearing design decision, not a formality.

- Single-snapshot rows (no per-vehicle time series), so sequence models (RNN/Transformer) are not a natural fit for this file.

3. Model Setup: Preprocessing & Architecture

Pipeline: stratified 70/15/15 train/val/test split (seed 42, stratified on Engine Condition to preserve the 63/37 ratio across splits) → StandardScaler fit on train only, applied to all splits → class-weighted loss (pos_weight in BCEWithLogitsLoss) to counteract the imbalance at the loss level. Gaussian noise jitter ($\sigma=0.05$, standardized units, train-only, applied fresh per mini-batch) is tested as an additional regularizer, given the dataset's modest size relative to a 6-feature nonlinear model — its effect is measured directly as an ablation in Section 4/5.5 rather than assumed.

Architecture (EngineConditionMLP, 993 parameters): tabular, non-sequential, non-spatial data (6 independent sensor readings per snapshot, no grid structure, no temporal ordering across rows) makes a fully-connected feedforward network the appropriate choice over a CNN or RNN.

Input(6, standardized)

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-> Linear(6->32) -> BatchNorm1d -> ReLU -> Dropout(0.3)
-> Linear(32->16) -> BatchNorm1d -> ReLU -> Dropout(0.2)
-> Linear(16->8) -> ReLU
-> Linear(8->1) [raw logit]
```

Design choice	Setting	Reasoning
Output / loss	Linear logit + BCEWithLogitsLoss(pos_weight)	Numerically stable sigmoid+BCE; pos_weight up-weights minority class
Hidden activation	ReLU	Avoids vanishing-gradient saturation; standard default
Optimizer	Adam, lr=1e-3, weight_decay=1e-4	Adaptive per-parameter rates; weight_decay adds an L2 smoothness prior
Batch size	32, mini-batch SGD	More gradient updates/epoch on a comparatively small (13.6k-row) train set
Dropout	0.3 / 0.2	Higher than a larger-data setting would need — smaller dataset raises overfitting risk per parameter
Early stopping	Patience 15 on val loss	Textbook overfitting countermeasure; training here converged in 39 epochs

4. Baselines vs. Final Model

Baseline: logistic regression (class_weight='balanced'), on the same scaled 6-feature input, plus a single-feature (Engine rpm only) variant to confirm no individual feature is close to sufficient. Four models were trained and evaluated on the held-out test set (n=2,931):

Model	AUC	Acc	Recall(0)	Recall(1)	F1(0)	F1(1)
LR — full features (baseline)	0.696	0.651	0.605	0.678	0.562	0.710
LR — Engine rpm only (baseline)	0.669	0.636	0.571	0.674	0.537	0.700
MLP — no augmentation (final model)	0.700	0.636	0.729	0.581	0.597	0.668
MLP — with Gaussian jitter (ablation)	0.698	0.628	0.729	0.568	0.592	0.658

Prior exploratory work on this dataset (see project history) additionally trained Random Forest (300 trees, AUC 0.685) and Gradient Boosting (300 trees, AUC 0.700, and AUC 0.700 again after adding 6 engineered interaction/ratio features) — all cluster within the same ~0.70 band, corroborating the ceiling discussed in Section 5.

Selection: the MLP without augmentation is the final model. It achieves the highest AUC (0.700) among the reproducible models and, more importantly given the imbalance, lifts class-0 recall from 60.5% (LR) to 72.9% — directly addressing the poor minority-class recall the baseline exposed — at the cost of class-1 recall (58.1% vs. 67.8%

for LR). Whether this trade-off is desirable depends on the unresolved label-semantics question (Section 6). The augmented variant is reported but not selected as final: it scores marginally lower on every metric (AUC 0.698 vs. 0.700), an honest negative ablation result discussed further in Section 5.5.

5. Evaluation

5.1 Metric Selection — What and Why

- Accuracy — reported, but explicitly not the headline metric: a trivial always-predict-class-1 model already scores 63% given the imbalance, while carrying zero discriminative value.
- Per-class Precision/Recall/F1 (not macro-averaged) — required for two reasons: macro-averaging would let the majority class mask minority-class performance, and the label-semantics ambiguity (Section 6) means either class could be the operationally important one, so both are shown symmetrically.
- Confusion matrix — makes the false-negative/false-positive split directly visible, which is the operationally relevant trade-off given the safety framing in Section 1 (a missed fault is worse than a false alarm).
- ROC-AUC — threshold-independent, so LR and MLP can be compared on equal footing regardless of where each model's decision threshold happens to sit.

5.2 Results

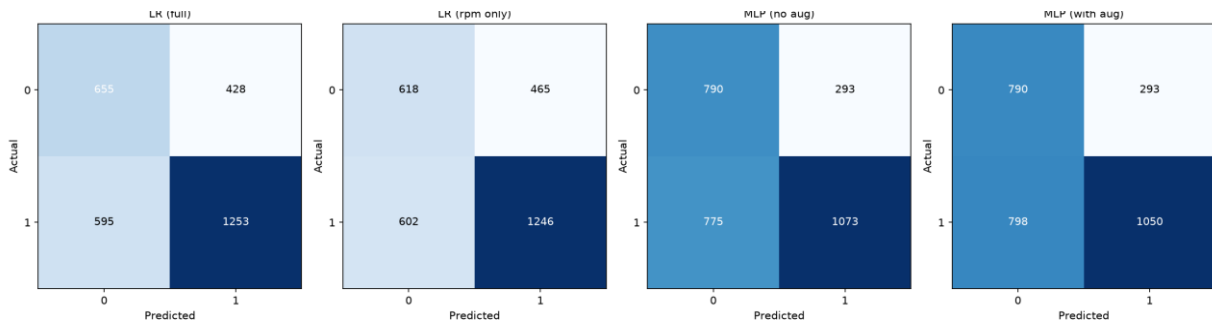


Figure 1. Confusion matrices, test set ($n=2,931$), all four models.

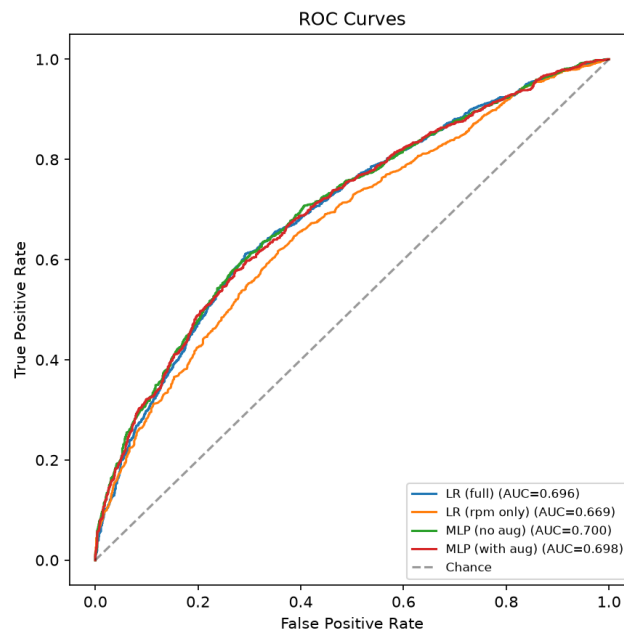


Figure 2. ROC curves, test set.

5.3 Failure Case: RPM-Band Accuracy

RPM quartile	RPM range	n	MLP accuracy
Q1	77–591	729	0.790
Q2	591–748	736	0.663
Q3	748–943	732	0.545
Q4	943–2191	734	0.545

MLP accuracy is far from uniform across the input space: it falls from 79.0% in the lowest RPM quartile to 54.5% in the highest — near chance level. The model's aggregate 63.6% test accuracy therefore hides a strong RPM-dependent failure mode, which any deployment threshold or confidence estimate would need to account for.

5.4 Interpretation: Permutation Importance

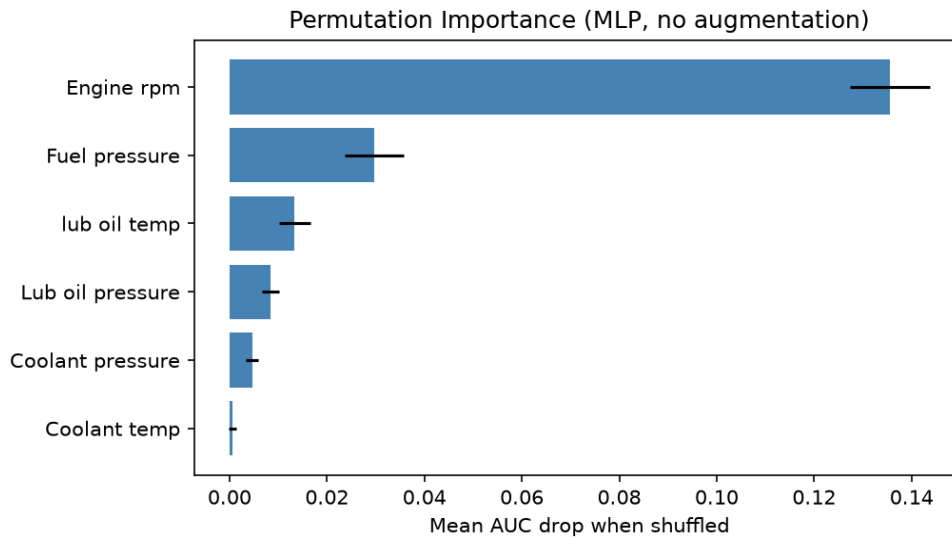


Figure 3. Mean AUC drop when each feature is shuffled (MLP, no augmentation, 10 repeats).

Engine rpm dominates (mean AUC drop 0.135) — roughly 4.6x the next-ranked feature, Fuel pressure (0.030). This is consistent with the correlation audit (Section 2), where Engine rpm was also the strongest individual correlate — the trained MLP concentrates decision-relevant signal on the same feature the linear audit flagged, rather than spreading importance broadly across all six inputs.

5.5 Augmentation Ablation

Gaussian jitter ($\sigma=0.05$, train-only) was tested as a regularizer and produced a small negative result: AUC 0.698 vs. 0.700 without it, with every other metric also slightly lower (Section 4 table). Reported honestly rather than omitted — the most likely explanation is that dropout, batch normalization, and weight decay already sufficiently regularize this 993-parameter model on 13,674 training rows, leaving no headroom for jitter to add value, and instead just adding noise the model has to fit around.

6. Limitations

- Label semantics unconfirmed: no documentation establishes whether Engine Condition=0 means healthy or faulty. This directly limits how the class-0/class-1 recall trade-off in Section 4 should be read — results are reported symmetrically rather than assuming a direction.
- Performance ceiling ($\sim$ AUC 0.70): confirmed as a data information ceiling, not a modeling shortfall — logistic regression, random forest, gradient boosting (with and without engineered features), and the MLP all cluster within a few points of 0.70 (Section 4). No amount of further architecture tuning on these 6 features is expected to move this materially.

- No documented data provenance or collection methodology — unclear whether this is real fleet telemetry, a bench rig, or partially synthetic; reported figures should be read as a methodology demonstration rather than a deployability claim.
- Only 6 sensor channels — real OBD-II systems expose more signal (throttle position, O2 sensor, battery voltage); a production system would likely have a higher achievable ceiling.
- Dataset size (19,535 rows) is modest for a 6-feature nonlinear model — mitigated by the regularization stack (Section 3) but still a real constraint on usable model capacity.
- Single-snapshot rows (no per-vehicle time series) rule out sequence models (RNN/Transformer) for this data, despite their relevance to the broader model-family comparison (Theoretical Q1).

7. Conclusion — Was This Data and Setup the Right Choice?

Largely yes, for the stated goal. The dataset's weak individual feature correlations and a clearly-beatable linear baseline (AUC 0.696–0.669) gave the MLP a genuine, non-trivial bar to clear — unlike an initially-considered alternative dataset where a single feature nearly solved the task outright. The MLP's small capacity (993 parameters, matched to a 6-feature input), class-weighted loss, and regularization stack were appropriate choices for a 19.5k-row, moderately-imbalanced tabular problem, and the augmentation ablation shows those regularization choices were validated empirically rather than assumed.

The setup's limitation is the data, not the model: six weakly-correlated sensor channels impose an $AUC \approx 0.70$ ceiling that held across every model family tested. Given the safety-relevant framing in Section 1, the honest conclusion is that this feature set alone is not sufficient for high-confidence deployment — the productive next step is more sensor channels or per-vehicle time-series data (which would motivate revisiting the RNN/Transformer family from Theoretical Q1), not further tuning of the current architecture.